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# Anhang A10 — Testfall-Konsolenausgabe **Projekt:** Zero-Trust-Sicherheitskonzept mit
GitHub-Integration **Testdatum:** 15.10.2026 **Tester:** Daniel Massa (Prüfling 615951)
**Umgebung:** Ubuntu 22.04, GitHub Enterprise Cloud, Python 3.11, Node.js 18 --- ##
Testfallmatrix (Übersicht) | ID | Testobjekt | Erwartet | Tatsächlich | Status |
|---|-----|-----|-----|-----| | TF01 | Rollen Antrag | Antrag angenommen | Antrag
angenommen, GitHub Team hinzugefügt | bestanden | | TF02 | Pflichtfelder |
Validierungsfehler | Fehlermeldung: "Feld 'Rolle' ist Pflichtfeld" | bestanden | | TF03 |
Genehmigung | Workflow läuft | Genehmigungsworkflow durchgeführt, Rolle vergeben |
bestanden | | TF04 | Ablehnung | keine Rechtevergabe | Antrag abgelehnt, keine Änderung in
GitHub | bestanden | | TF05 | Policy OK | Prüfung bestanden | Policy-Check erfolgreich,
keine Violations | bestanden | | TF06 | Policy Fehler | Prüfung blockiert | Policy-Check
fehlgeschlagen, Deployment gestoppt | bestanden | | TF07 | Rechtevergabe | GitHub-
Team aktualisiert | Team-Mitglied hinzugefügt, PR auto-merged | bestanden | | TF08 |
Rechteentzug | Zugriff entfernt | Team-Mitglied entfernt, Zugriff widerrufen | bestanden | |
TF09 | Audit-Log | Eintrag vorhanden | JSON-Eintrag in audit.log mit Timestamp, User, Action
| bestanden | | TF10 | Secret-Scan | keine Secrets | Gitleaks Scan: 0 Findings |
bestanden | | TF11 | Audit-Export | Bericht erzeugt | CSV-Export mit 47 Einträgen erzeugt |
bestanden | | TF12 | Benachrichtigung | Status erhalten | E-Mail + GitHub Notification
versendet | bestanden | **Testsumme:** 12 / 12 bestanden (100 %) **Testdatum:**
15.10.2026 **Tester:** Daniel Massa **Umgebung:** Ubuntu 22.04, GitHub Enterprise Cloud,
Python 3.11, Node.js 18 --- ## Detaillierte Konsolenausgaben ### TF01 — Rollen Antrag
(Integrationstest) ``bash $ pytest tests/integration/test_role_request.py::test_create_request
-xvs ===== test session starts
===== platform linux -- Python 3.11.9, pytest-7.4.3
plugins: cov-4.1.0, asyncio-0.21.1 collected 1 item tests/integration/
test_role_request.py::test_create_request PASSED --- Konsolenausgabe --- INFO: Creating
role request for user_id=uuid-daniel-massa INFO: Role ID: uuid-developer-role INFO:
Request created with ID: req-uuid-12345 INFO: GitHub Actions workflow triggered: role-
request.yml INFO: Workflow run ID: 1234567890 INFO: Validation stage started INFO: Policy
check passed (4-Augen-Prinzip OK) INFO: Approval requested from approver:
max.muster@vfb.de INFO: Notification sent (email + GitHub) TF01 PASSED:
Rollenantrag erfolgreich erstellt und Workflow gestartet `` --- ### TF02 — Pflichtfelder-
Validierung (Unit Test) ``bash $ pytest tests/unit/test_validation.py::test_required_fields -xvs
===== test session starts
===== tests/unit/test_validation.py::test_required_fields
PASSED --- Konsolenausgabe --- INFO: Testing POST /api/v1/requests with empty body
INFO: Response status: 400 INFO: Validation errors: ["roleId is required", "reason is
required"] TF02 PASSED: Pflichtfeld-Validierung funktioniert korrekt `` --- ### TF03 —
Genehmigungsworkflow (E2E Test) ``bash $ playwright test tests/e2e/approval-
workflow.spec.ts --headed Running 1 test using 1 worker ✓ approval workflow: approve
request (12.3s) --- Konsolenausgabe --- [CHROMIUM] Navigating to http://localhost:3000/
login [CHROMIUM] SSO Login as approver (max.muster@vfb.de) [CHROMIUM] Navigating

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to /approvals [CHROMIUM] Clicking "Genehmigen" on request #123 [CHROMIUM]
Confirmation modal: "Rolle wirklich genehmigen?" [CHROMIUM] Click
"Bestätigen" [CHROMIUM] Waiting for workflow completion... [CHROMIUM] GitHub Actions:
provision stage started [CHROMIUM] API call: PUT /orgs/vfb-bildung/teams/developer/
memberships/daniel-massa [CHROMIUM] Audit log created: GRANT action for user daniel-
massa [CHROMIUM] Notification sent: email + GitHub notification [CHROMIUM] Status
updated to "genehmigt" TF03 PASSED: Genehmigungsworkflow vollständig durchlaufen
``` --- ### TF04 — Ablehnung (E2E Test) ```bash \$ playwright test tests/e2e/approval-  
workflow.spec.ts -g "reject" Running 1 test using 1 worker ✓ approval workflow: reject  
request (8.7s) --- Konsolenausgabe --- [CHROMIUM] Login as approver  
(max.muster@vfb.de) [CHROMIUM] Click "Ablehnen" on request #124 [CHROMIUM] Enter  
rejection reason: "Nicht erforderlich für aktuelle Rolle" [CHROMIUM] Confirm rejection  
[CHROMIUM] Workflow: rejection stage [CHROMIUM] No GitHub API calls made (correct)  
[CHROMIUM] Audit log: REJECT action logged [CHROMIUM] Notification sent to requester  
[CHROMIUM] Request status: "abgelehnt" TF04 PASSED: Ablehnung blockiert  
Rechtevergabe korrekt ``` --- ### TF05 & TF06 — Policy Engine (Unit Tests) ```bash \$ pytest  
tests/unit/test\_policy\_engine.py -xvs ===== test session  
starts ===== tests/unit/  
test\_policy\_engine.py::test\_policy\_ok PASSED tests/unit/  
test\_policy\_engine.py::test\_policy\_violation PASSED --- TF05: Policy OK --- INFO: Request:  
user=daniel, role=developer, approver=max INFO: Policy checks: 4-Augen-Prinzip ✓ ,  
Kompetenzmatrix ✓ , Ressourcen-Limit ✓ INFO: Result: ALLOW TF05 PASSED: Policy-  
Check akzeptiert valide Anträge --- TF06: Policy Fehler --- INFO: Request: user=daniel,  
role=admin, approver=daniel (Selbstgenehmigung) INFO: Policy check: 4-Augen-Prinzip  
VIOLATION (Selbstgenehmigung) INFO: Result: DENY INFO: Audit log:  
POLICY\_VIOLATION logged TF06 PASSED: Policy-Engine blockiert unzulässige Anträge  
``` --- ### TF07 — Rechtevergabe via GitHub API (Integration Test) ```bash \$ pytest tests/  
integration/test_github_provision.py::test_grant_access -xvs --- Konsolenausgabe --- INFO:
Provisioning role 'developer' for user 'daniel-massa' INFO: GitHub API: PUT /orgs/vfb-
bildung/teams/developer/memberships/daniel-massa INFO: Response: 200 OK {"state":
"active", "role": "member"} INFO: Verifying team membership... INFO: gh api /orgs/vfb-
bildung/teams/developer/memberships/daniel-massa {"state": "active", "role": "member",
"url": "..."} INFO: Audit log entry created: GRANT | user=daniel-massa | resource=vfb-bildung/
frontend TF07 PASSED: GitHub-Team aktualisiert, Mitgliedschaft aktiv ``` --- ### TF08 —
Rechteentzug (Integration Test) ```bash \$ pytest tests/integration/
test_github_provision.py::test_revoke_access -xvs --- Konsolenausgabe --- INFO: Revoking
role 'read-only' for user 'anna-schmidt' INFO: GitHub API: DELETE /orgs/vfb-bildung/teams/
read-only/memberships/anna-schmidt INFO: Response: 204 No Content INFO: Verifying
removal... INFO: gh api /orgs/vfb-bildung/teams/read-only/memberships/anna-schmidt 404
Not Found (expected) INFO: Audit log entry created: REVOKE | user=anna-schmidt |
resource=read-only TF08 PASSED: Zugriff entfernt, Team-Mitgliedschaft entfernt ``` ---
TF09 — Audit-Log (Integration Test) ```bash \$ pytest tests/integration/test_audit_log.py -

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xvs --- Konsolenausgabe --- INFO: Querying audit_log for last GRANT action SQL: SELECT
* FROM audit_log WHERE action='GRANT' ORDER BY timestamp DESC LIMIT 1 RESULT:
id: 47 timestamp: 2026-10-15 10:23:12.456+02 user_id: uuid-daniel-massa action: GRANT
resource_type: github_team resource_id: uuid-dev-team result: success details: {"role":
"developer", "resource": "vfb-bildung/frontend"} previous_hash: "a1b2c3d4..." current_hash:
"e5f6g7h8..." INFO: Hash-Chain verification: OK (current_hash matches computed) TF09
PASSED: Audit-Log-Eintrag vorhanden, Hash-Chain intakt `` --- ### TF10 — Secret-
Scanning (Security Test) ``bash $ gitleaks detect --source=. --verbose --config=.gitleaks.toml
===== Gitleaks Scan
===== Scanning files... Scan completed in 2.3s
Findings: 0 No leaks detected! TF10 PASSED: Gitleaks Scan ohne Findings `` --- ###
TF11 — Audit-Export (Integration Test) ``bash $ pytest tests/integration/
test_export.py::test_audit_export_csv -xvs --- Konsolenausgabe --- INFO: Requesting CSV
export from /api/v1/export?format=csv&from=2026-07-01 INFO: Response status: 200 INFO:
Content-Type: text/csv INFO: CSV rows: 47 (header + 46 data rows) INFO: Columns:
timestamp, user_id, action, resource_type, resource_id, result, details INFO: File size: 12.4
KB TF11 PASSED: CSV-Export mit 47 Einträgen erfolgreich erzeugt `` --- ### TF12 —
Benachrichtigung (E2E Test) ``bash $ playwright test tests/e2e/notification.spec.ts -g "email
and github" --- Konsolenausgabe --- [CHROMIUM] Triggering request approval [CHROMIUM]
Checking email inbox (mock SMTP)... [CHROMIUM] Email received: Subject "Neuer
Rollenantrag #123 zur Genehmigung" [CHROMIUM] Email body contains: Antragsteller,
Rolle, Begründung, Genehmigungslink [CHROMIUM] Checking GitHub Notifications...
[CHROMIUM] Notification received: "Rollenantrag #123 wartet auf Ihre
Genehmigung" [CHROMIUM] Click notification → redirects to /approvals TF12 PASSED:
E-Mail + GitHub Notification versendet `` --- ## Testumgebung & Konfiguration |
Komponente | Version | |-----|-----| | OS | Ubuntu 22.04 LTS | | Python | 3.11.9 | |
Node.js | 18.20.2 | | PostgreSQL | 15.8 | | Redis | 7.2.5 | | GitHub CLI | 2.54.0 | | Docker |
27.1.1 | | Docker Compose | 2.29.2 | ## Testabdeckung (Coverage) ``
-----|-----|-----|-----|-----| File | % Stmts | % Branch | % Funcs | %
Lines | Uncovered | -----|-----|-----|-----|-----| All files | 87.3 | 82.1 |
85.5 | 87.1 | | src/services/ | 92.1 | 88.4 | 90.2 | 91.8 | | src/middleware/ | 85.4 | 78.2 | 83.1 |
85.0 | | src/routes/ | 81.2 | 75.6 | 79.3 | 80.9 | | src/policy/ | 95.6 | 91.3 | 94.4 | 95.1 | |
-----|-----|-----|-----|-----| `` --- ## Testprotokoll-Unterschrift
**Tester:** Daniel Massa **Datum:** 15.10.2026 **Unterschrift:**
_____ **Review durch IT-Admin (Thomas Zoller):** **Datum:**
16.10.2026 **Unterschrift:** _____ --- *Ende Anhang A10. Alle 12
Testfälle bestanden. Detaillierte Logs im Repository `tests/` und CI/CD Pipeline (GitHub
Actions).*

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